COMPSCI 689
Lecture 8: Convex Non-Differentiable Optimization

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Support Vector Classification

$$f_\theta(\mathbf{x}) = \mathrm{sign}(g_\theta(\mathbf{x}))$$

$$g_\theta(\mathbf{x}) = \mathbf{x}\theta = \mathbf{x}\mathbf{w} + b$$

$$\hat{\theta} = \arg\min_{\theta} C \sum_{n=1}^{N} \max(0, 1 - y_n g_\theta(\mathbf{x}_n)) + ||\mathbf{w}||_2^2$$

Global Optimality of Convex Non-Differentiable Functions

- Our existing optimization theory only holds for the case of differentiable functions.
- However, it turns out that many results generalize to the case of non-differentiable functions that are convex and we can use them to directly minimize convex non-differentiable regularized risk functions.
- To begin, strongly convex non-differentiable functions have a unique global minima, exactly as with strongly convex differentiable functions.
- We will begin with the characterization of the minimizer of a non-differentiable convex function. [1]

[1] This section mostly follows the presentation in Boyd's EE364b - Convex Optimization II course.

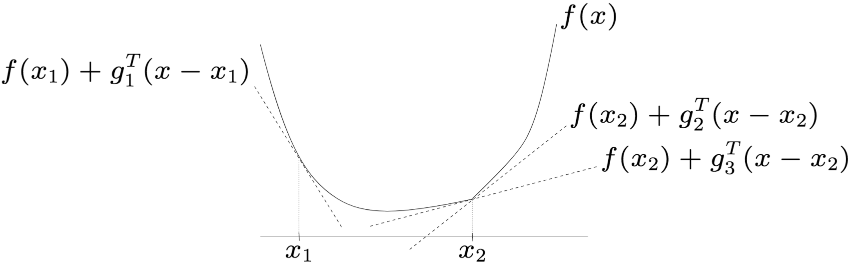
Subgradient

- Let $f : \mathbb{R}^D \rightarrow \mathbb{R}$. A vector $\mathbf{g} \in \mathbb{R}^D$ is said to be a sub-gradient of f at a point $\mathbf{x}_o \in \mathbb{R}^D$ if for all $\mathbf{x} \in \mathbb{R}^D$:

$$f(\mathbf{x}) \geq f(\mathbf{x}_o) + \mathbf{g}^T \cdot (\mathbf{x} - \mathbf{x}_o)$$

- That is to say, the hyperplane defined by $h(\mathbf{x}) = f(\mathbf{x}_o) + \mathbf{g}^T \cdot (\mathbf{x} - \mathbf{x}_o)$ lies at or below below f everywhere and touches f at $\mathbf{x}_o$.

Example: Subgradients

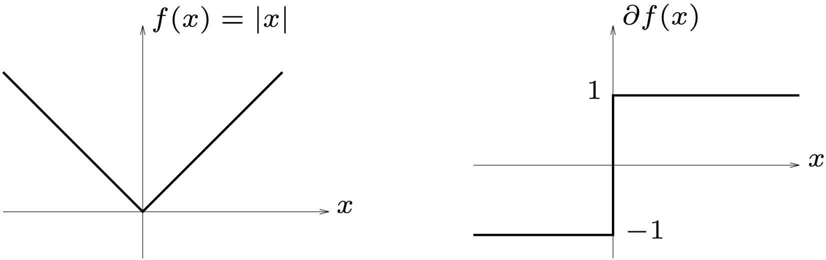


In this example, $\mathbf{g}_1$ is the unique subgradient of f at $\mathbf{x}_1$. Due to f being non-differentiable at $\mathbf{x}_2$, both $\mathbf{g}_2$ and $\mathbf{g}_3$ are subgradients of f at $\mathbf{x}_2$.

Subdifferentials

- If f is convex and is differentiable at $\mathbf{x}_o$, then $\nabla f(\mathbf{x}_o)$ is its unique subgradient at $\mathbf{x}_o$.
- If f is convex and non-differentiable at $\mathbf{x}_o$, it will generally have more that one vector $\mathbf{g}$ satisfying the subgradient property.
- The set of all subgradients of f at $\mathbf{x}_o$ is called the subdifferential of f at $\mathbf{x}_o$ denoted by $\partial f(\mathbf{x}_o)$.
- $\partial f(\mathbf{x}_o)$ is a closed, convex set in $\mathbb{R}^D$. If f is convex, $\partial f(\mathbf{x}_o)$ is always non-empty.

Example: Subdifferentials



The righthand plot shows $\partial f(x)$ for $f(x) = |x|$.
We have $\partial f(\mathbf{x}) = \{\text{sign}(x)\}$ for $x \neq 0$.
When $x = 0$, the line $|0| + g \cdot (x - 0) = g \cdot x$ will lie below f everywhere only if $g \in [-1, 1]$.

Characterizing the Global Minimum

- Let $f : \mathbb{R}^D \to \mathbb{R}$ be a convex function.
- $\mathbf{x}_*$ is the global minimizer of f if and only if $\mathbf{0} \in \partial f(\mathbf{x}_*)$.
- This is a generalization of the idea of a stationary point to include the case of non-differentiable functions.

Sub-gradient Descent

- If f is differentiable at $\mathbf{x}$ then $\partial f(\mathbf{x}) = \{\nabla f(\mathbf{x})\}$ and $-\mathbf{g} = -\nabla f(\mathbf{x})$ is a descent direction unless $\nabla f(\mathbf{x}) = 0$.
- If f is not differentiable at $\mathbf{x}$, and $\mathbf{0} \notin \partial f(\mathbf{x})$, then there exists at least one $\mathbf{g} \in \partial f(\mathbf{x})$ where $-\mathbf{g}$ is a descent direction.
- If f is not differentiable at $\mathbf{x}$, then there may exist some $\mathbf{g} \in \partial f(\mathbf{x})$ where $-\mathbf{g}$ is a **not** descent direction.

Sub-gradient Descent

Despite the fact that not all sub-gradients yield descent directions, it is still possible to minimize a convex non-differentiable function using a fairly basic sub-gradient descent procedure:

- Initialize $\mathbf{x}_0 \in \mathbb{R}^D, f_{min} = \infty, \mathbf{x}_* = 0$
- For k from 1 to K:
  - Let $\mathbf{g}_k \in \partial f(\mathbf{x}_{k-1})$
  - Set $\mathbf{x}_k = \mathbf{x}_{k-1} - \alpha_k \mathbf{g}_k$
  - If $f(\mathbf{x}_k) < f_{min}$ then set $f_{min} = f(\mathbf{x}_k)$ and $\mathbf{x}_* = \mathbf{x}_k$
- Return $\mathbf{x}_*$

Questions: How to choose α_k? How to choose K?

Subgradient Descent Convergence

- Line search is typically not used for sub-gradient descent procedures. It is more common to use a fixed sequence of step sizes.
- Common step size rules include $\alpha_k = \alpha/(\beta + k)$ or $\alpha_k = \alpha/\sqrt{k}$.
- Under either of these step size rules, we have that the sequence of subgradient descent iterates $\mathbf{x}_k$ satisfies:

$$\lim_{k\to\infty} f(\mathbf{x}_k) = \min_{\mathbf{x}} f(\mathbf{x})$$

Finding Subdifferentials

- Suppose that x_0 is a point of non-differentiability for a 1-dimensional convex function $f(x)$.
- Suppose that in the neighborhood $[a, x_0]$, for some $a < x_0$, the value of $f(x)$ is given by a differentiable function $g(x)$ and in the neighborhood $[x_0, b]$ for some $b > x_0$, the value of $f(x)$ is given by a differentiable function $h(x)$.
- Then, the subdifferential of $f(x)$ at x_0 is:

$$\partial f(x_0) = \left[\frac{dg(x)}{dx}|_{x_0}, \frac{dh(x)}{dx}|_{x_0} \right]$$

Partial Linearity and Chain Rule

The subdifferential operator satisfies partial linearity and chain rule properties:

- **Scaling:** If $f(\mathbf{x})$ is a convex function and $\alpha > 0$, then if $\mathbf{g} \in \partial f(\mathbf{x})$, $\alpha\mathbf{g} \in \partial(\alpha f(\mathbf{x}))$
- **Addition:** If $f_1(\mathbf{x})$ and $f_2(\mathbf{x})$ are convex functions and $\mathbf{g}_1 \in \partial f_1(\mathbf{x})$ and $\mathbf{g}_2 \in \partial f_2(\mathbf{x})$, then $\mathbf{g}_1 + \mathbf{g}_2 \in \partial(f_1(\mathbf{x}) + f_2(\mathbf{x}))$
- **Restricted Chain Rule**: Let $f : \mathbb{R} \to \mathbb{R}$ be a convex function, $f' : \mathbb{R} \to \mathbb{R}$ provide a sub-gradient of f for any input, and $h(\mathbf{x}) = \mathbf{x}\mathbf{a} + b$ be a linear function. Then, $f'(h(\mathbf{x}))\nabla h(\mathbf{x}) = f'(h(\mathbf{x}))\mathbf{a}^T \in \partial f(h(\mathbf{x}))$.

These properties can be used to determine the subdifferentials of more complex functions if they reduce to certain combinations or compositions of simpler functions.

Example: Finding a Subgradient for SVC

- We begin with the risk function:

$$\begin{aligned} R(\theta, \mathcal{D}) &= C \sum_{n=1}^{N} L(y_n g_\theta(\mathbf{x}_n)) + ||\mathbf{w}||_2^2 \\ &= C \sum_{n=1}^{N} \max(0, 1 - y_n g_\theta(\mathbf{x}_n)) + ||\mathbf{w}||_2^2 \end{aligned}$$

- For any setting of the parameters θ We are looking for a vector $\mathbf{h}$ such that: $\mathbf{h} \in \partial R(\theta, \mathcal{D})$

Example: Finding a Subgradient for SVC

- By the additivity and scaling properties, we have that if $\mathbf{h}_n \in \partial L(y_n, g_\theta(\mathbf{x}_n))$ and $\mathbf{h}_w \in \partial ||\mathbf{w}||_2^2$, then:

$$C \sum_{n=1}^{N} \mathbf{h}_n + \mathbf{h}_w \in \partial R(\theta, \mathcal{D})$$

- We will proceed by finding suitable vectors $\mathbf{h}_n$ and $\mathbf{h}_w$ using properties of sub-gradients.

Example: Finding a Subgradient for SVC

- Consider $\mathbf{h}_w \in \partial||\mathbf{w}||_2^2$.
- Since $||\mathbf{w}||_2^2$ is a differentiable function with gradient $2\mathbf{w}$, we have that $\mathbf{h}_w = 2[\mathbf{w}; 0]$.

Example: Finding a Subgradient for SVC

- Consider $\mathbf{h}_n \in \partial L(y_n, g_\theta(\mathbf{x}_n))$.
- Recall $L(y_n, g_\theta(\mathbf{x}_n)) = \max(0, 1 - y_n, g_\theta(\mathbf{x}_n))$ and $g_\theta(\mathbf{x}_n) = \mathbf{x}_n\theta$
- It is more convenient to analyze this function as $\ell(y_n g_\theta(\mathbf{x}_n)) = \max(0, 1 - y_n, g_\theta(\mathbf{x}_n))$
- In this case $\ell(z) = \max(0, 1 - z)$.
- By the restricted chain rule, we have the result below where $\ell'(z)$ is a sub-gradient function of the hinge loss:

$$\mathbf{h}_n = \ell'(y_n g_\theta(\mathbf{x}_n)) y_n \mathbf{x}_n^T \in \partial L(y_n, g_\theta(\mathbf{x}_n))$$

- We next need to figure out what $\ell'(z)$ should be ...

Example: Finding a Subgradient for SVC

- Consider $\partial\ell(z) = \partial \max(0, 1-z)$.
- This is a non-differentiable convex function with one point of non-differentiability at $z = 1$.
- For $z > 1$ the function is constant so $\partial\ell(z) = \{0\}$ for $z > 1$
- For $z < 1$ the function is $1 - z$ so $\partial\ell(z) = \{-1\}$ for $z < 1$.
- At $z = 0$ we have $\partial\ell(z) = [-1, 0]$ following subdifferential rules for 1-D convex functions.
- We choose the following sub-gradient function for the hinge loss:

$$\ell'(z) = \begin{cases} 0 & \dots\ z \geq 1 \\ -1 & \dots\ z < 1 \end{cases}$$

Example: Finding a Subgradient for SVC

- This gives us a final answer for a subgradient of the risk at the current parameters:

$$\mathbf{h} = C\sum_{n=1}^{N} \mathbf{h}_n + \mathbf{h}_w \quad (1)$$

$$= C\sum_{n=1}^{N} \ell'(y_n\mathbf{x}_n\theta)y_n\mathbf{x}_n^T + 2[\mathbf{w};0] \quad (2)$$

- This subgradient can be passed in to a sub-gradient descent algorithm in order to learn SVC model parameters.